Hacettepe University - Computer Engineering
CMP720 Embedded System Design - Spring 2026

Final Project Topic Proposal Template

(Fill in the sections below. Replace placeholders with your content.)

Group Members (Name Surname + Student ID)

Instructions: List every group member as "Name Surname — Student ID".

- **Ali Bölücü — N25123069**

Project Topic Title

Instructions: Provide a concise, descriptive title.

Spectral-Seeded NSGA-II for Multi-Application Mapping in Many-Core Systems

Project Track (choose one)

❑ **Implementation Track**

■ **Optimization Track**

Abstract (300–500 words)

Instructions: In 300–500 words, explain: (i) the problem tackled, (ii) the proposed solution approach and methodology, (iii) high-level system architecture and major components, and (iv) an initial plan and milestones for attacking the problem. Include one crude block diagram (schematic for Implementation track / optimization pipeline diagram for Optimization track) to illustrate the idea.

(i) Problem: In modern Many-Core systems and Multiprocessor Systems-on-Chip (MPSoC), simultaneously mapping multiple applications onto a Network-on-Chip (NoC) architecture is a complex NP-hard problem. Existing multi-objective evolutionary algorithms, such as NSGA-II, typically rely on random initial populations. In massive search spaces, this leads to slow convergence and sub-optimal results, as the algorithm often gets trapped in local optima regarding communication cost, latency, and thermal distribution.

(ii) Proposed Solution: This project proposes integrating a "Spectral Partitioning" and Clustering pre-processing stage into the initialization phase of the NSGA-II-based mapping model. Our methodology relies on converting the communication task graphs of multiple incoming applications into a unified Affinity Matrix. By calculating the Laplacian of this matrix and performing eigenvalue decomposition (Fiedler vector analysis), highly communicative tasks will be dynamically clustered together using the Silhouette coefficient. These optimized clusters obtained from the spectral analysis will be fed into the initial population of NSGA-II as "elite seeds" to significantly narrow down the search space.

(iii) System Architecture: The developed architecture will consist of three main components:

- *Data Parsing and Matrix Module: Takes the source, target, and weight vectors of applications to construct isolated compound graph matrices.*
- *Spectral Pre-processor: Merges highly communicative blocks by performing spectral bisection and topology clustering through mathematical matrix transformations.*
- *NSGA-II Algorithm: Combines randomly generated individuals with spectrally optimized individuals.*

(iv) Initial Plan and Milestones:

- *Step 1: Coding the baseline NSGA-II model and establishing the evaluation infrastructure.*
- *Step 2: Developing the Spectral Clustering module using Python/MATLAB to perform Laplacian and Fiedler vector calculations.*
- *Step 3: Integrating the pre-processing module into the initialization phase of the evolutionary algorithm.*
- *Step 4: Conducting tests on standard benchmark graphs and comparing the results with similiar researches in the field.*

Motivation / Societal Impact (200–500 words)

Instructions: In 200–500 words, explain why this problem matters for the greater good (societal impact, safety, sustainability, accessibility, cost reduction, reliability, etc.). Include who benefits and how you would measure impact.

When tasks on a computer chip are placed closer to each other, they communicate much more efficiently. This simple change reduces the workload on the chip's network and significantly lowers overall energy consumption. Additionally, our method prevents certain areas of the chip from overheating. Keeping the temperature balanced helps the hardware last longer and run more reliably. Ultimately, the practical success of this project will be measured by its ability to deliver three clear improvements over traditional methods. Specifically, we aim to achieve lower energy usage for every data packet, faster data transfer times across the chip, and a significantly quicker process for finding the best possible task arrangement.

Related Work / References

Add a few related works from the literature, patents, or existing products to show what other researchers and developers have done in this area (3–5 works are sufficient, but you may include more). At this stage, you are not expected to conduct a full literature review; that will be done after your project idea is accepted. ***Add entries in BibTeX format to `references.bib` and cite them in your text using `\cite{<bibtexhandle>}`****. The References will be ordered and generated automatically. You don't need to order them inside `references.bib` or delete references that you do not cite.*

- A New Solution based on Multi-objective Algorithm for Multi-application Mappings for Many-Core Systems [1].
- A spectral clustering approach to application-specific Network-on-Chip synthesis [2].
- Reinforcement Guided Genetic Algorithm for Application Mapping in Network-on-Chip Architectures: Toward Transparent and Efficient MPSoC Scheduling [3].
- ET: A Metaheuristic Optimization Algorithm for Task Mapping in Network-on-Chip [4].

References

[1] M. de Almeida, E. Pedrino, and I. Gallon, "A new solution based on multi-objective algorithm for multi-application mappings for many-core systems," *IEEE Latin America Transactions*, vol. 23, no. 4, pp. 323–328, 2025.

[2] V. Todorov, D. Mueller-Gritschneder, H. Reinig, and U. Schlichtmann, "A spectral clustering approach to application-specific network-on-chip synthesis," in *2013 Design, Automation Test in Europe Conference Exhibition (DATE)*, pp. 1783–1788, 2013.

[3] A. A. J. Al-Hchaimi, M. A. Al-Shareeda, A. T. Azar, and W. El-Shafai, "Reinforcement guided genetic algorithm for application mapping in network-on-chip architectures: Toward transparent and efficient mpsoc scheduling," *IEEE Access*, vol. 13, pp. 177520–177536, 2025.

[4] K. Li, J. Shao, and Y. Song, "Et: A metaheuristic optimization algorithm for task mapping in network-on-chip," *Electronics*, vol. 14, no. 14, 2025.

How to Submit

- Open the Google Drive sheet (submission tracker): `https://docs.google.com/spreadsheets/d/17ACM2EVBZ9tZY3XIKveWwRSbt-QfVjS7qkYcTWC3GwU/edit?usp=sharing`
- Fill in your group and project details in an appropriate row (make sure you do not delete/edit other students' entries!).
- Create a repository that you will use to submit **all** project work throughout the semester (GitHub or GitLab).
- Upload/commit your proposal files to the repository (both the `.zip` source files with `.tex` and the compiled `.pdf`).
- Paste the repository link into the corresponding column in the sheet.
- If your repository is private, add the instructor as a collaborator: `selmadilek@gmail.com`.